

Project Initialization and Planning Phase

Date	15 March 2024
Team ID	xxxxxxx
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Create a problem statement to understand the challenges faced by users in predicting flood risks. This template helps identify the needs, goals, and pain points of disaster management authorities, meteorologists, and people living in flood-prone regions. A well-defined problem statement enables the development of an accurate and reliable machine learning-based flood prediction system that supports early warning, informed decision-making, and effective disaster preparedness.

Example:

I am  Disaster management officer	I'm trying to  Predict flood risk in advance to issue timely warnings	But  Traditional forecasting methods are often inaccurate or delayed	Because  Large volumes of weather data require accurate predictive models	Which makes me feel  Concerned about public safety and disaster response.
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Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A disaster management officer responsible for public safety.	Predict flood risk in advance to issue timely warnings.	Existing forecasting methods do not provide timely or sufficiently accurate flood predictions.	They rely on traditional techniques that cannot efficiently analyze large volumes of weather and rainfall data.	Concerned about the safety of people and the effectiveness of disaster response.
PS-2	A resident living in a flood-prone area.	Know whether there is a possibility of flooding in my locality	I do not receive reliable and timely flood predictions.	Existing warning systems are limited, delayed, or difficult to access.	Anxious about the safety of my family, home, and property during heavy rainfall.